

Lab5: Building an Online Shopping Website

Objective

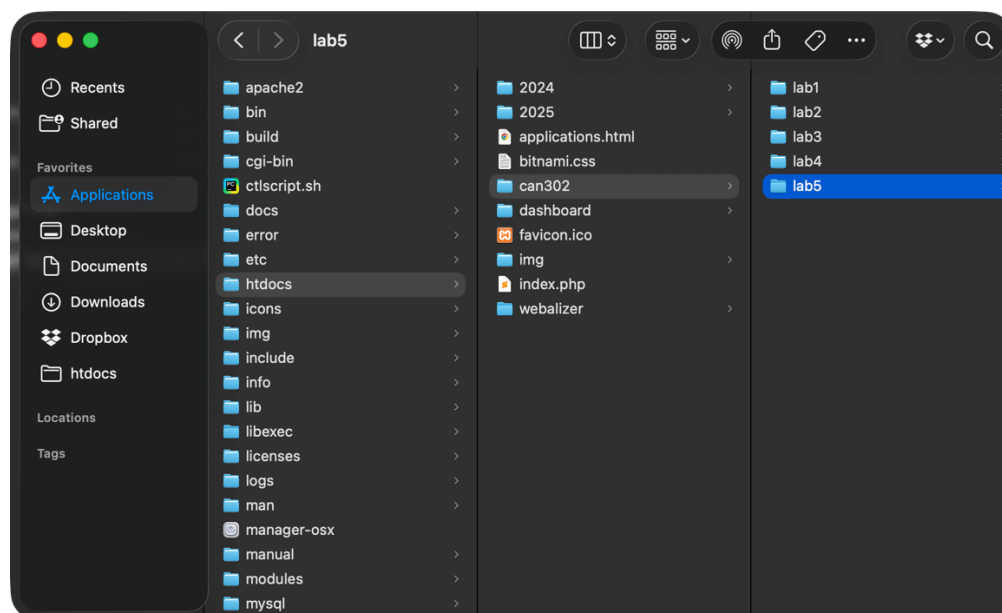
- Design and implement a static webpage using HTML, CSS, and Bootstrap
- Use JavaScript for basic interactivity (e.g., form validation, dynamic content updates)
- Integrate a MySQL database with the website using PHP
- Build a functional online shopping website with user authentication, product display, etc.

Tips

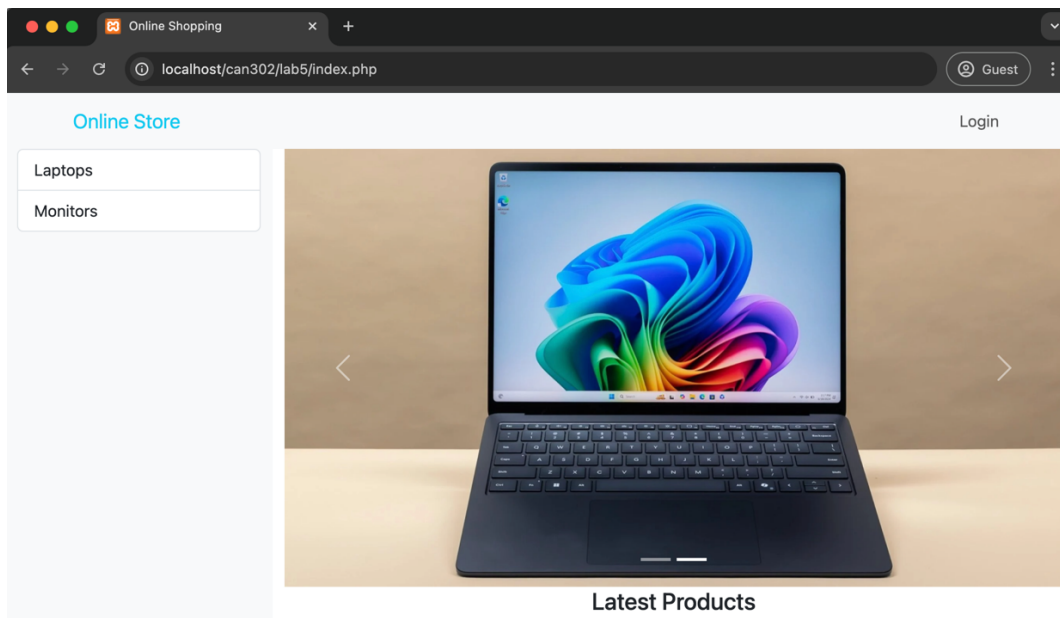
- These labs are expected to take more than the allocated **4 hours (Week 5 and Week 6)**. You should attempt the tasks before the lab. Practice makes perfect.
- Before the lab, you could post your questions in LMO forum, and you are welcome to offer solutions to the questions in LMO.
- During the lab, you could discuss with TAs for your questions. Face to face discussion will have a relatively high efficiency.

A. Static Webpage Design (Week 5)

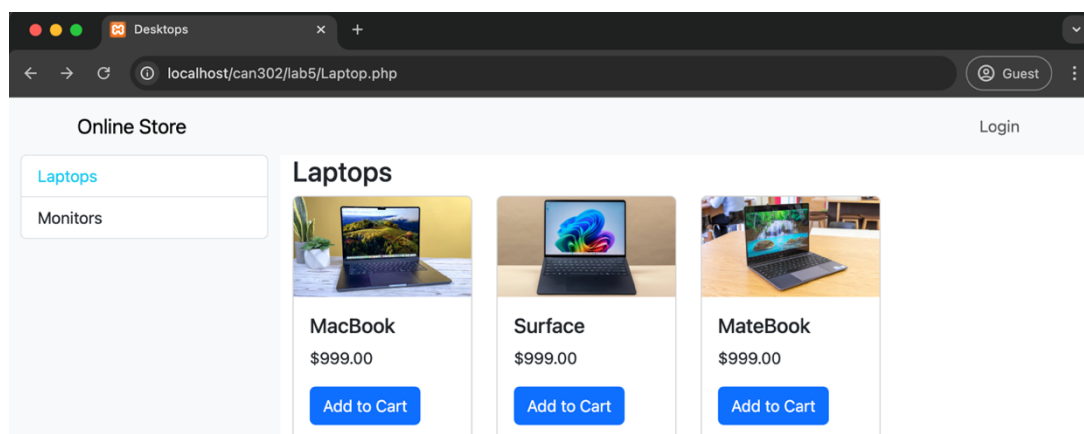
1. Create a new subfolder "lab5" under folder "can 302" (The following screenshot is in macOS, but same operations can be applied to Windows):



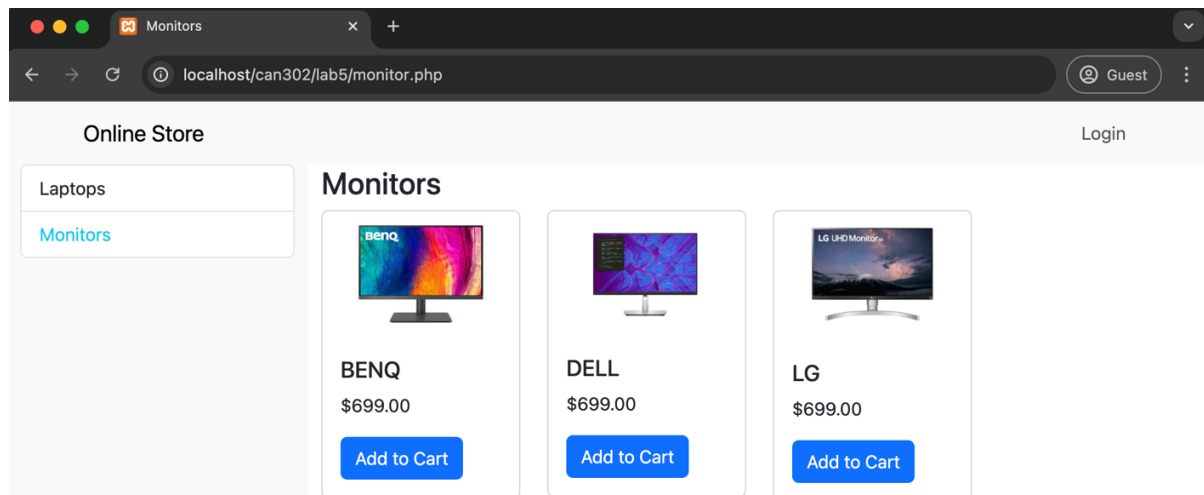
2. Create a subfolder "img" under "lab5"
3. Download provided images and save in "img"
4. Create a new file "index.php" under "lab5", and design a webpage like:



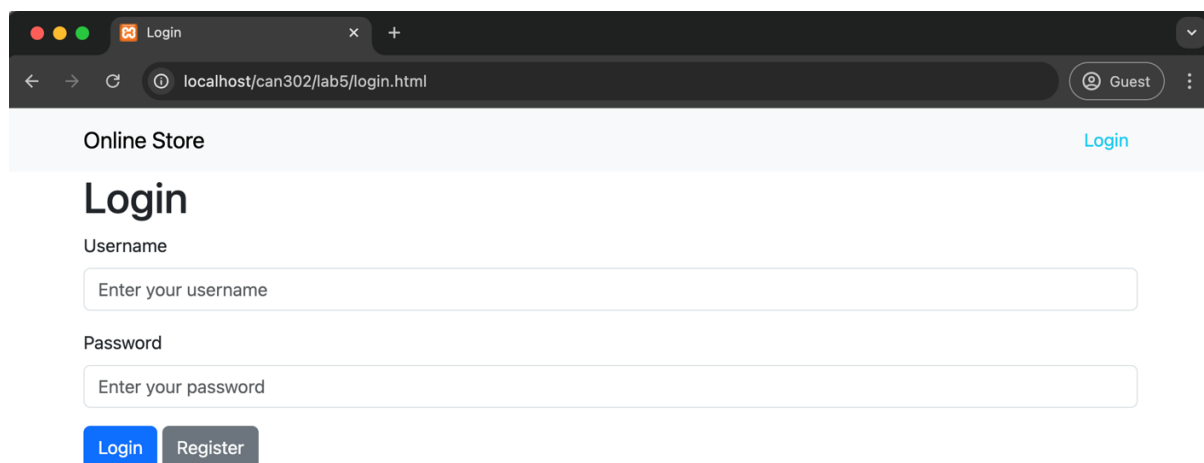
- (a) Top navigation menu includes two buttons "Online Store" and "Login" (use Bootstrap's "navbar" component, please refer to Lab3-B9)
 - (b) Left sidebar includes two categories of "Laptops" and "Monitors" (use Bootstrap's "list-group" component, see <https://getbootstrap.com/docs/5.3/components/list-group/>)
 - (c) Carousel displays two hot products with images (images are in "img", use Bootstrap's "carousel" component, please refer to Lab3-B13)
5. Create a new file "laptop.php" under "lab5", and design a webpage as follows when click the "Laptops" button:



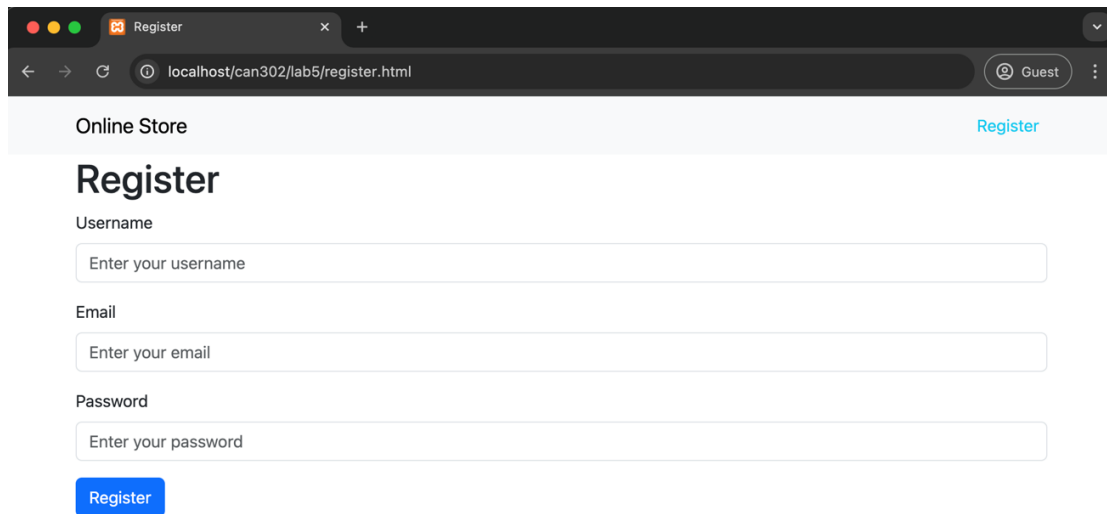
- (a) Include the "name" and "price" for each product
 - (b) Include an "Add to Cart" button for each product
 - (c) Use Bootstrap's "card" component, see <https://getbootstrap.com/docs/5.3/components/card/>
6. Similarly, create a new file "monitor.php" under "lab5", and design a webpage as follows when click the "Monitor" button:



7. Create a new file "login.html" under "lab5", and design a webpage as follows when click the "Login" button:

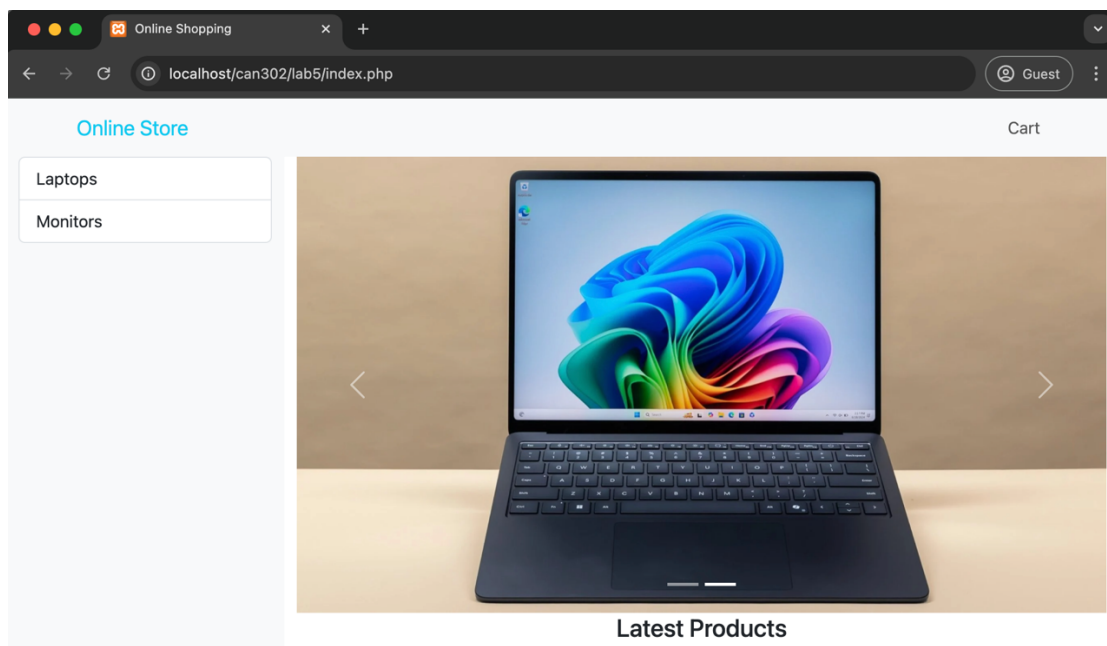


- (a) Create a form with fields for "Username" and "Password"
 - (b) Include a "Login" button and a "Register" button
 - (c) Use Bootstrap's "form" component, please refer to Lab4-A20
8. Similarly, create a new file "regiser.html" under "lab5", and design a webpage as follows when click the "Register" button in "login.html":



The screenshot shows a web browser window with the address bar displaying 'localhost/can302/lab5/register.html'. The page has a dark header with 'Online Store' on the left and a 'Register' link on the right. The main content area is titled 'Register' and contains three input fields: 'Username' with the placeholder 'Enter your username', 'Email' with the placeholder 'Enter your email', and 'Password' with the placeholder 'Enter your password'. Below these fields is a blue 'Register' button.

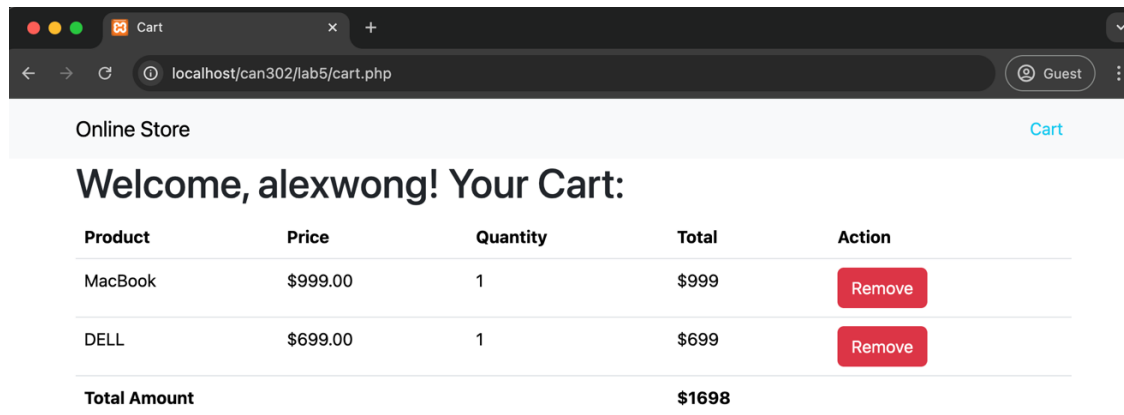
9. If you have already registered an account, you can login and the homepage should change to:



- (a) The "login" button changes to "Cart"
- (b) Use PHP to display dynamic content
- (c) Sample code in the "navbar" component:

```
<div class="collapse navbar-collapse" id="navbarNav">
  <ul class="navbar-nav ms-auto">
    <?php
      if (isset($_SESSION['username'])) {
        echo '<li class="nav-item"><a class="nav-link" href="cart.php">Cart</a></li>';
      } else {
        echo '<li class="nav-item"><a class="nav-link" href="login.html">Login</a></li>';
      }
    <?>
  </ul>
</div>
```

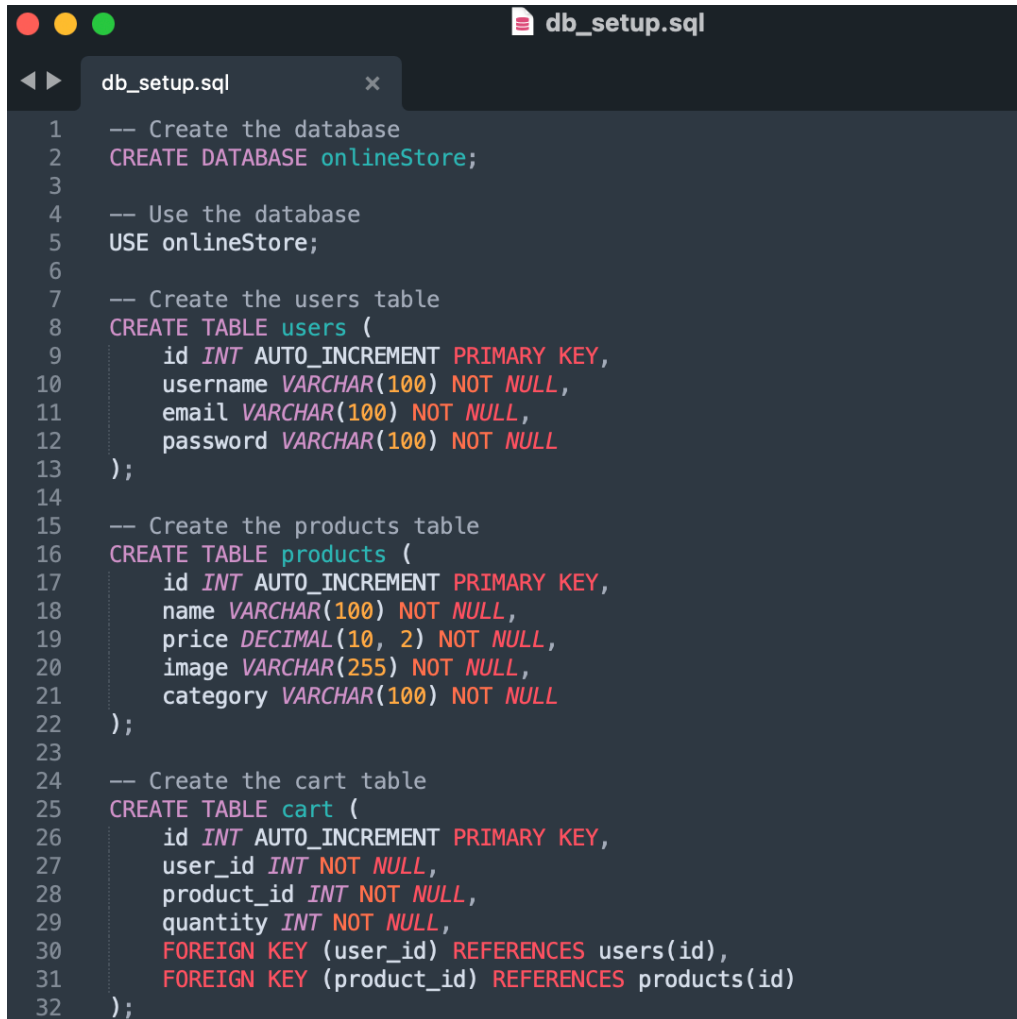
10. Create a new file "cart.php" under "lab5", and design a webpage as follows when click the "Cart" button in "index.php":



- (a) Include "Product", "Price", "Quantity", and "Total" for each product
- (b) Include a "Remove" button for each product
11. In this part of the lab session, we focus on designing **static webpages** without interacting with a database. Use your knowledge from previous lab sessions and Bootstrap's documentation to create an online shopping website. Particularly, Bootstrap's utility classes are utilized for layout, spacing, and responsiveness, though custom CSS can be added if needed. By the end of this session, we have created the following webpages:
- (a) "index.php" includes a top navigation menu, a left sidebar, and a carousel
- (b) "laptop.php" includes three products, each with an image, name, and price
- (c) "monitor.php" includes three products, each with an image, name, and price
- (d) "login.html" includes a form with two fields for "Username" and "Password"
- (e) "register.html" includes a form with three fields for "Username", "Email", and "Password"
- (f) "cart.php" includes a table with columns for "Product", "Price", "Quantity", and "Total"

B. Database Integration (Week 6)

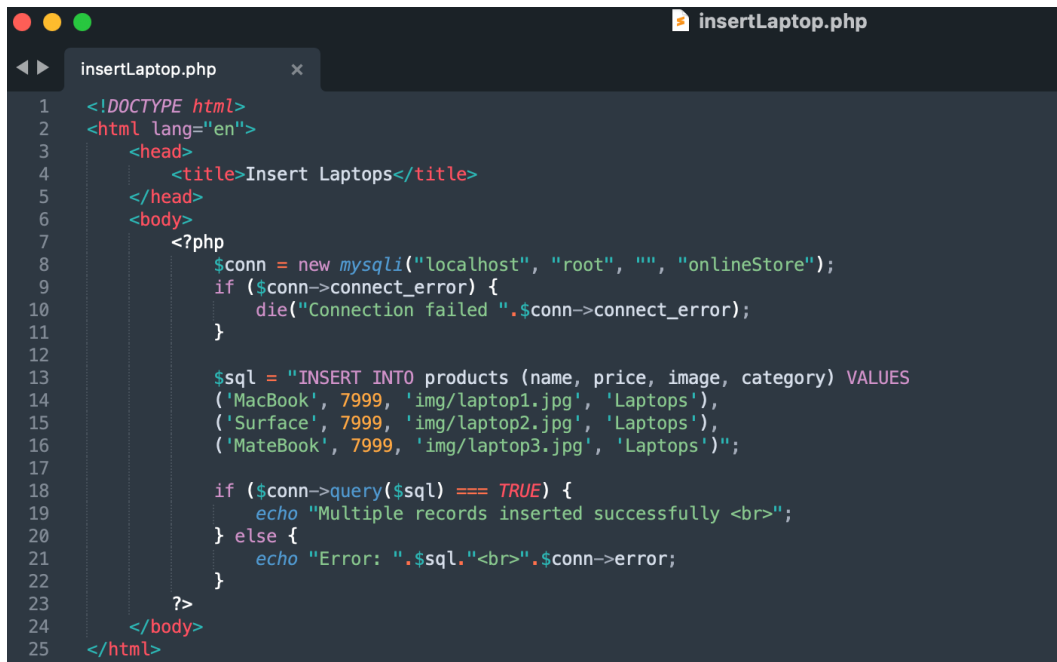
1. Download the provided "db_setup.sql" file, and import it into <http://localhost/phpmyadmin/> to create a database "onlineStore" along with three tables: "users", "products", and "cart". The SQL file is as follows:



```
1  -- Create the database
2  CREATE DATABASE onlineStore;
3
4  -- Use the database
5  USE onlineStore;
6
7  -- Create the users table
8  CREATE TABLE users (
9      id INT AUTO_INCREMENT PRIMARY KEY,
10     username VARCHAR(100) NOT NULL,
11     email VARCHAR(100) NOT NULL,
12     password VARCHAR(100) NOT NULL
13 );
14
15 -- Create the products table
16 CREATE TABLE products (
17     id INT AUTO_INCREMENT PRIMARY KEY,
18     name VARCHAR(100) NOT NULL,
19     price DECIMAL(10, 2) NOT NULL,
20     image VARCHAR(255) NOT NULL,
21     category VARCHAR(100) NOT NULL
22 );
23
24 -- Create the cart table
25 CREATE TABLE cart (
26     id INT AUTO_INCREMENT PRIMARY KEY,
27     user_id INT NOT NULL,
28     product_id INT NOT NULL,
29     quantity INT NOT NULL,
30     FOREIGN KEY (user_id) REFERENCES users(id),
31     FOREIGN KEY (product_id) REFERENCES products(id)
32 );
```

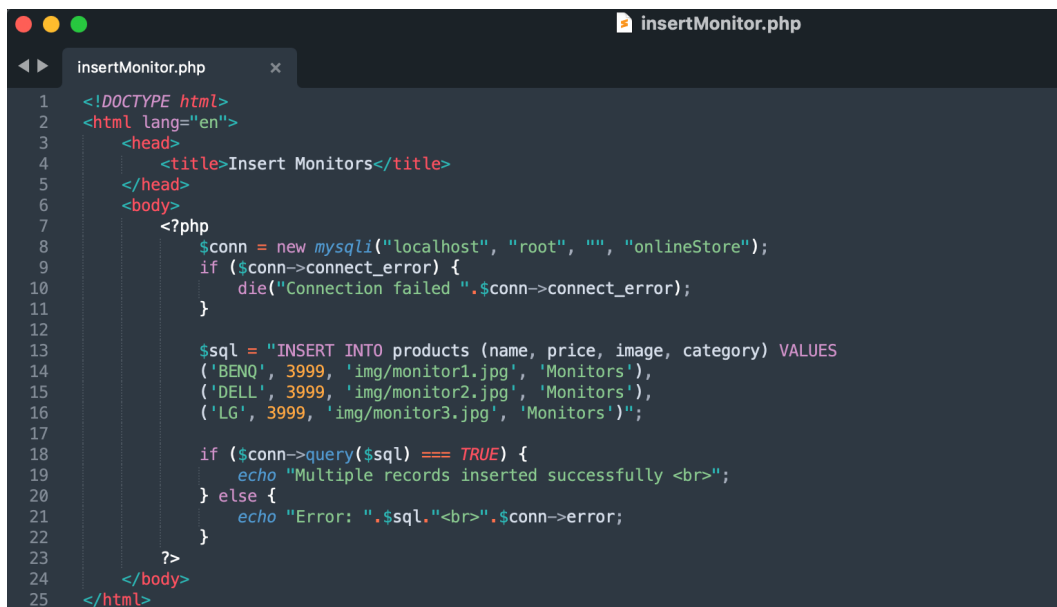
- (a) If there already exists a database named "onlineStore", you can choose another name, or delete the current one with **"DROP DATABASE onlineStore"** and then import the above file
- (b) The "users" table has 4 columns: "id" (primary key, auto increment), "username", "email", and "password"
- (c) The "products" table has 5 columns: "id" (primary key, auto increment), "name", "price", "image", and "category"
- (d) The "cart" table has 4 columns: "id" (primary key, auto increment), "user_id", "product_id", and "quantity"

2. Insert multiple records of "Laptops" into the "products" table as follows:



```
1 <!DOCTYPE html>
2 <html lang="en">
3   <head>
4     <title>Insert Laptops</title>
5   </head>
6   <body>
7     <?php
8       $conn = new mysqli("localhost", "root", "", "onlineStore");
9       if ($conn->connect_error) {
10         die("Connection failed ".$conn->connect_error);
11       }
12
13       $sql = "INSERT INTO products (name, price, image, category) VALUES
14         ('MacBook', 7999, 'img/laptop1.jpg', 'Laptops'),
15         ('Surface', 7999, 'img/laptop2.jpg', 'Laptops'),
16         ('MateBook', 7999, 'img/laptop3.jpg', 'Laptops')";
17
18       if ($conn->query($sql) === TRUE) {
19         echo "Multiple records inserted successfully <br>";
20       } else {
21         echo "Error: ".$sql."<br>".$conn->error;
22       }
23     <?>
24   </body>
25 </html>
```

- (a) Visit <http://localhost/can302/lab5/insertLaptop.php> to insert data into the table
- (b) The category for the inserted records is "Laptops", as shown in the code above
3. Similarly, insert multiple records of "Monitors" into the "products" table as follows:



```
1 <!DOCTYPE html>
2 <html lang="en">
3   <head>
4     <title>Insert Monitors</title>
5   </head>
6   <body>
7     <?php
8       $conn = new mysqli("localhost", "root", "", "onlineStore");
9       if ($conn->connect_error) {
10         die("Connection failed ".$conn->connect_error);
11       }
12
13       $sql = "INSERT INTO products (name, price, image, category) VALUES
14         ('BENQ', 3999, 'img/monitor1.jpg', 'Monitors'),
15         ('DELL', 3999, 'img/monitor2.jpg', 'Monitors'),
16         ('LG', 3999, 'img/monitor3.jpg', 'Monitors')";
17
18       if ($conn->query($sql) === TRUE) {
19         echo "Multiple records inserted successfully <br>";
20       } else {
21         echo "Error: ".$sql."<br>".$conn->error;
22       }
23     <?>
24   </body>
25 </html>
```

- (a) Visit <http://localhost/can302/lab5/insertMonitor.php> to insert data into the table
- (b) The category for the inserted records is "Monitors", as shown in the code above
4. Now that we have created the necessary database and tables for this lab session, the next step is to modify the previously created webpages to read and update data from database.

- At the beginning of "index.php", add PHP code to start a session using "session_start()" and connect to the database as follows:



```

1 <?php
2 session_start();
3 $conn = new mysqli("localhost", "root", "", "onlineStore");
4 if ($conn->connect_error) {
5     die("Connection failed " . $conn->connect_error);
6 }
7 ?>

```

- Revise the "carousel" part in "index.php" to fetch images from database as follows:



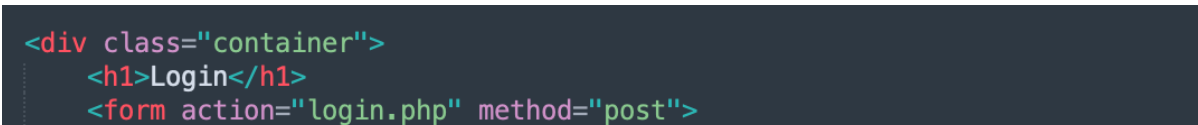
```

<!-- Carousel -->
<div class="col-md-9">
    <div id="carouselDemo" class="carousel slide" data-bs-ride="carousel">
        <div class="carousel-indicators">
            <button type="button" data-bs-target="#carouselDemo" data-bs-slide-to="0" class="active" aria-current="true"></button>
            <button type="button" data-bs-target="#carouselDemo" data-bs-slide-to="1"></button>
        </div>
        <div class="carousel-inner">
            <?php
                $get_slides = "SELECT * FROM products LIMIT 1";
                $run_slides = mysqli_query($conn, $get_slides);
                $row_slides = mysqli_fetch_array($run_slides);
                echo '<div class="carousel-item active"></div>';

                $get_slides = "SELECT * FROM products LIMIT 1 OFFSET 1";
                $run_slides = mysqli_query($conn, $get_slides);
                while($row_slides = mysqli_fetch_array($run_slides)){
                    echo '<div class="carousel-item"></div>';
                }
            ?>
        </div>
        <button class="carousel-control-prev" type="button" data-bs-target="#carouselDemo" data-bs-slide="prev">
            <span class="carousel-control-prev-icon"></span>
        </button>
        <button class="carousel-control-next" type="button" data-bs-target="#carouselDemo" data-bs-slide="next">
            <span class="carousel-control-next-icon"></span>
        </button>
    </div>
    <div class="container">
        <div class="text-center">
            <h4>Latest Products</h4>
        </div>
    </div>
</div>

```

- Include two images in the carousel
 - See: https://www.w3schools.com/mysql/mysql_limit.asp
- Revise the "laptop.php" to display products by fetching data from "products" table
 - Revise the "monitor.php" to display products by fetching data from "products" table
 - For "login.html", we should use the "post" method in the form as follows:



```

<div class="container">
    <h1>Login</h1>
    <form action="login.php" method="post">

```

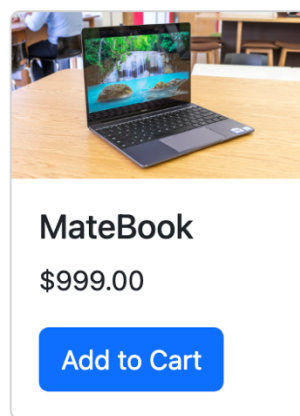
- Thus, we need to create a new file named "login.php" (refer to Lab4-A21). Besides, add code to check if the user exists in the "users" table. If the user exists, redirect to the "cart.php"; otherwise, redirect to the "register.html"

11. For "register.html", we should also use the "post" method in the form as follows:

```
<div class="container">
  <h1>Register</h1>
  <form action="register.php" method="post">
```

12. Thus, we need to create a new file named "register.php" (refer to Lab4-C3). Besides, start a session using "session_start()" (refer to Lab4-A24) and redirect to the "cart.php"

13. For each product shown in "laptop.php" and "monitor.php", there is a "Add to Cart" button as the following example, and we can add this product to the cart by clicking this button. Thus, we create a new file named "addCart.php":



- (a) Start a session using "session_start()"
- (b) Check if user is logged in; if not, redirect to the "login.html"
- (c) Connect to the database
- (d) Check if the product is in the cart; if yes, increase the quantity by 1
- (e) If the product is not in the cart, insert a new record with the quantity set to 1

14. For each product listed in "cart.php", there is a "Remove" button as the following example, and we can remove this product from the cart by clicking this button. Thus, we create a new file named "removeCart.php":

DELL	\$699.00	1	\$699	<button>Remove</button>
------	----------	---	-------	-------------------------

- (a) Start a session using "session_start()"
- (b) Check if user is logged in; if not, redirect to the "login.html"
- (c) Connect to the database
- (d) Remove the record from database

15. In this part of the lab session, we focus on integrating the MySQL database with the webpages created using PHP. Practice establishing connections between the PHP code and the MySQL database. Write PHP functions to query the database for product data to display on webpages.

By the end of this session, we have modified/created the following webpages:

- (a) "index.php"
- (b) "laptop.php"
- (c) "monitor.php"
- (d) "login.php"
- (e) "register.php"
- (f) "cart.php"
- (g) "addCart.php"
- (h) "removeCart.php"

Summary

In the two lab sessions, we successfully developed a dynamic online shopping website using HTML, CSS, JavaScript, Bootstrap, PHP, and SQL. Key accomplishments include:

- Front-End Development: Designed a responsive interface using HTML, CSS, and Bootstrap. Implemented features such as a carousel to showcase products
- Back-End Integration: Created a MySQL database for user information, product details, and cart data. Utilized PHP to establish connections between the front-end and database
- Core Functionalities: Implemented key e-commerce features, including dynamic product display, user authentication (login/register), and cart management (add/remove items)